

Video-of-Thought

Step-by-step video reasoning from perception to cognition

Hao Fei, Shengqiong Wu, Wei Ji, Hanwang Zhang, Meishan Zhang, Mong Li Lee, Wynne Hsu

Outline

Problem & Objectives

- Motivation: Limits of Prior Approaches
- Solution: MotionEpic + Video-of-Thought

Scalability and Reliability

- VoT Reasoning Walkthrough
- Pipeline, Evaluation, and Takeaways

Video Reasoning Gap

Perception and cognition stay disconnected over time.

- Leads to target drift and temporal-causal errors.

Targeted Capabilities

Ground + track targets
Stable relations across time
Commonsense actions + context
Step-by-step reasoning
Accurate, verifiable answers

Why Methods Fail

Brittle grounding in dynamics

Shallow temporal + causal reasoning

Errors cascade without grounding-aware steps

Breadth & Efficiency

Scale-out breadth: cross-dataset, zero-shot; best acc.

- Scale-up efficiency: strong results with leaner tuning.

Practical: incremental updates; light, rare retraining.

- Modeling choices and tuning targets improve efficiency.

Model	Acc@D	Acc@E	Acc@P			Acc@C		
			A	R	AR	A	R	AR
• SoTA baselines								
TranSTR	73.6	75.8	65.1	65.0	48.9	68.6	65.3	50.3
Video-LLaMA	69.2	71.0	63.6	62.4	44.4	65.4	60.1	45.0
VideoChat	72.9	73.9	65.2	63.1	45.9	66.0	62.7	45.8
Video-ChatGPT	73.1	75.1	66.0	63.9	46.0	67.8	63.6	50.0
Video-LLaVA	73.7	74.4	67.6	65.4	47.7	68.0	64.9	51.5
• CoT								
Video-LLaVA	74.2	74.8	68.0	65.7	48.1	70.3	65.7	52.9
Video-LLaVA+stsg	75.7	75.9	68.9	67.2	50.0	70.7	67.2	53.6
MotionEpic	78.5	77.2	70.1	70.8	52.4	71.2	69.1	55.0
• VoT								
MotionEpic	81.2	83.0	74.3	73.7	54.7	74.5	73.8	58.6

Fast tuning light retrain

MotionEpic + VoT

MotionEpic builds STSGs: objects, relations, time.

- VoT: set task, track targets, analyze, rank+verify.

STSG Guided Grounding

STSG grounds objects and relations robust to occlusions

- Temporal links keep IDs across views; prevent cascades

Model	MSR-VTT	ActivityNet	NEXt-QA	STAR	AVG.
• Zero-shot SoTA baselines					
InternVideo	-	-	49.1	41.6	-
Video-LLaMA	49.6	21.4	43.5	36.4	37.7
VideoChat	52.0	26.5	52.8	45.0	44.1
Video-ChatGPT	54.3	35.2	53.0	48.7	47.8
Video-LLaVA	59.2	45.3	57.3	50.6	53.1
VideoChat2	54.1	49.1	61.7	59.0	56.0
• CoT					
Video-LLaVA	60.0	46.9	59.5	52.0	54.6
Video-LLaVA+STSG	61.5	48.4	60.6	52.7	55.8
MotionEpic	63.1	50.0	61.9	56.5	57.8
• VoT					
MotionEpic	66.2	54.6	66.5	61.7	62.3
w/o Verify-G	63.6	51.4	62.0	59.1	59.0
w/o Verify-C	65.1	53.4	62.8	58.8	60.1

Cross-dataset, zero-shot.

Context-Aware Actions

- STSG infers actions in relational and temporal context.
- Anticipates outcomes; adapts targets as scenes change.

VoT Reasoning Steps

- 1 Task + STSG targets -> 2 Tracking -> 3 Commonsense
- 4 Rank answers by evidence -> 5 Verify with STSG Trace.

End-to-End Pipeline

- Frames → STSG ground. → VoT reason. → verified answers
- Describe, explain, predict, counterfactual QA

Results & Takeaways

MotionEpic+VoT: SOTA; strong zero-shot

Ablations: each step counts; removals hurt

STSG + VoT => reliable, scalable